

Chintan Shah

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SUMMARY

Biomedical Engineer specializing in the end-to-end development of precision medical devices, with direct industry experience in intraocular lens (IOL) mechanical testing and process validation at Alcon. Proficient in SolidWorks CAD, Design Controls, and Verification & Validation (V&V) within ISO 13485. Adept at cross-functional collaboration, rigorous SOP documentation, and translating complex user requirements into patent-pending, clinically validated medical device solutions.

SKILLS AND COMPETENCIES

- **Design & Prototyping:** SolidWorks (CAD), DfX Principles, 3D Printing (FDM), Rapid Prototyping, Mechanism Design, Fixture Design.
- **Quality & Regulatory:** ISO 13485, GxP Compliance, SOPs, Design Controls, FMEA/Risk Management, CAPA, EU MDR, cGMP.
- **Validation & Testing:** V&V Testing, Design Verification, IQ/OQ/PQ Protocols, Test Method Validation (TMV), Bench Testing, Gage R&R.
- **Technical Tools & Data:** Python, Minitab (SPC/Six Sigma DMAIC), GraphPad Prism, InSpec (Micro Vu), Statistical Analysis, Root Cause Analysis.
- **Communication Skills:** Technical Writing & Documentation, Continuous Improvement Facilitation, Problem Solving

PROFESSIONAL EXPERIENCE

Robotic Production Associate, *Matic Robots, Sunnyvale, CA*

Mar 2026 – Present

- Facilitate **process validation and quality assurance** for robotic hardware and autonomous cleaning systems, ensuring all units correspond to established performance and durability standards before deployment in a high-volume production setting.
- Perform **functional testing and sensor calibration** for camera-based navigation stacks, identifying, documenting, and resolving hardware-software deviations to maintain system integrity.
- Manage the final inspection and shipping procedures for consumer units, using stringent quality control checkpoints to guarantee all goods match set performance criteria before release.

Research Associate, *Yaron Lab, Tempe, AZ*

Jan 2025 – Feb 2026

- Developed a patent-pending medical equipment for the **TAT2 endoscopic ink delivery system** by designing and 3D-printing two universal syringe attachments (screw-advance and spring-plunger) utilizing **FDM printing with PLA**.
- Iterated through multiple **design concepts**, evaluating mechanical trade-offs between the screw-advance and spring-plunger mechanisms across ergonomics, one-handed operability, and volume control precision to select the lead design.
- Evaluated both designs against a conventional syringe across three injection conditions with six volunteers, demonstrating that the spring-plunger produced significantly reduced volume variance ($p < 0.0001$) and the most consistent delivery across all tests.
- Conducted **ex vivo tests on porcine intestinal tissue**, demonstrating that the ink marking area with the spring-plunger was roughly 46% less than that with a normal syringe ($p = 0.023$), hence validating enhanced spot containment.
- Utilized **GraphPad Prism** to analyze experimental data and created a custom ImageJ/FIJI macro to quantitatively gauge the ink spread area and contrast across tissue samples.

R&D Process Intern, Alcon, *Belmont, CA*

May 2025 – Aug 2025

- Conducted precise **bench testing and Test Method Validation (TMV)** to assess essential mechanical specifications (axial stiffness and haptic height) for **Intraocular Lenses (IOL)**, directly contributing to the **design verification** of advanced ophthalmic devices.
- Developed a data analysis method in Python to characterize axial stiffness data and derive haptic height for Intraocular Lenses; conducted **Gage R&R** studies in **Minitab** to validate measurement system reliability.
- Developed **InSpec (Micro Vu)** scripts to automate quality inspection workflows, transitioning IOL batch testing from sample-based to full 100% inspection of 40-unit batches, resulting in enhanced compliance and increased throughput.
- Designed and 3D-printed mechanical **model eye** to support Train-the-Trainer sessions for laser activation procedures, replicating clinical ocular geometry to serve as a functional training prop.

Research Intern, Rapid Manufacturing Lab (IIT Bombay), *Mumbai, India*

Dec 2019 – Mar 2020

- Developed **mechanical prototypes** using SolidWorks and AutoCAD while applying Finite Element Analysis (FEA) to reduce modeling errors by 20% and optimize structural performance.
- Fabricated functional components by **CNC machining and additive manufacturing** to enable structural reliability testing for intricate engineering assemblies.
- Conducted structural reliability testing and recorded results to facilitate iterative development stages for intricate engineering assemblies.

EDUCATION

Master of Science in Biomedical Engineering | Arizona State University

Bachelor of Technology in Biomedical Engineering | University of Mumbai (Dwarkadas. J. Sanghvi College of Engineering)